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2장. 프로그래밍 언어의 역사

Programming Languages

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학습 내용

- Fortran
- Cobol
- Algol
- Pascal
- C
- C++
- C#

Fortran

Fortran

- Formular translation language
- First high-level programming language
- 1954
- IBM, John Backus

Fortran

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Versions

FORTRAN V 1962	FORTRAN 77 ANSI	FORTRAN 90	...

특징

- ❖ 효율성 (= 빠른 실행)
- ❖ 정적 (static) 메모리 관리

Fortran

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Sample

```
C       THIS IS SAMPLE PROGRAM
        PROGRAM MAIN
        INTEGER SUM
        READ (5, 100) K
100     FORMAT(I5)
        SUM = 0
        DO 200 I = 1, K
        SUM = SUM + I
200     CONTINUE
        PRINT *, 'SUM=', SUM
        STOP
        END
```

Cobol

Cobol

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- Common business oriented language
- Grace Hopper
- 1959 ~ 1960

특징

- 낮은 가독성
- Record 도입
- 선언부 (declaration) + 실행부 (execution)

Cobol

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- History
  - CODASYL: 1959 - The Conference on Data Systems Language
  - ANSI Cobol: 1968
  - Cobol-74
  - Cobol-85
  - Cobol 2002: ISO, object-orientation concepts

Cobol

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- 중요성
  - 비즈니스 관련 시스템 개발에 많이 이용되었음
  - 과거 잘 나가던 Cobol, ~ - https://www.cwn.kr/news/articleView.html?idxno=15667
  - there is noting that can replace COBOL - https://www.techspot.com/news/101900-face-there-nothing-can-replace-cobol.html

Cobol

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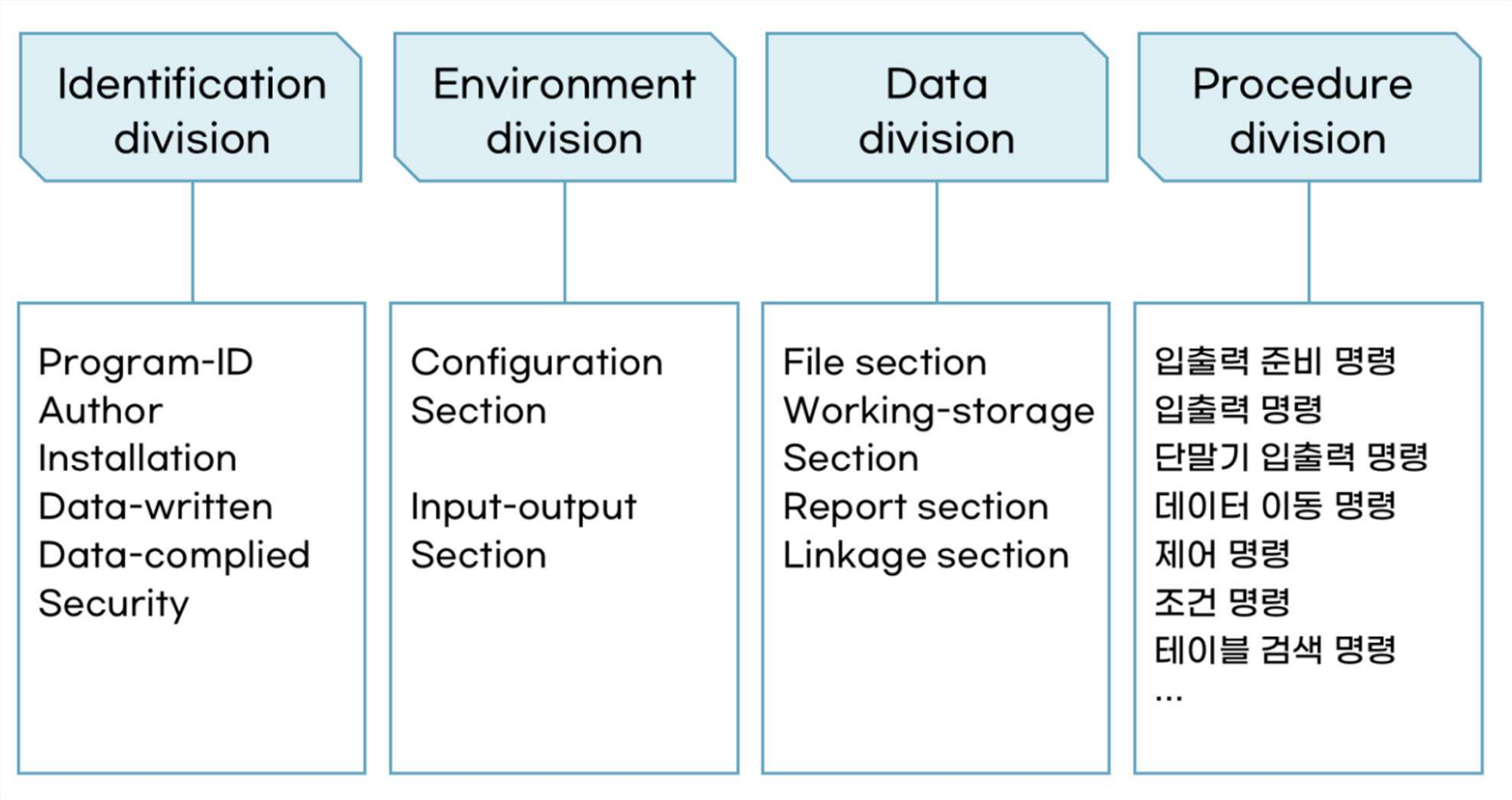
Division (부)

Section (절)

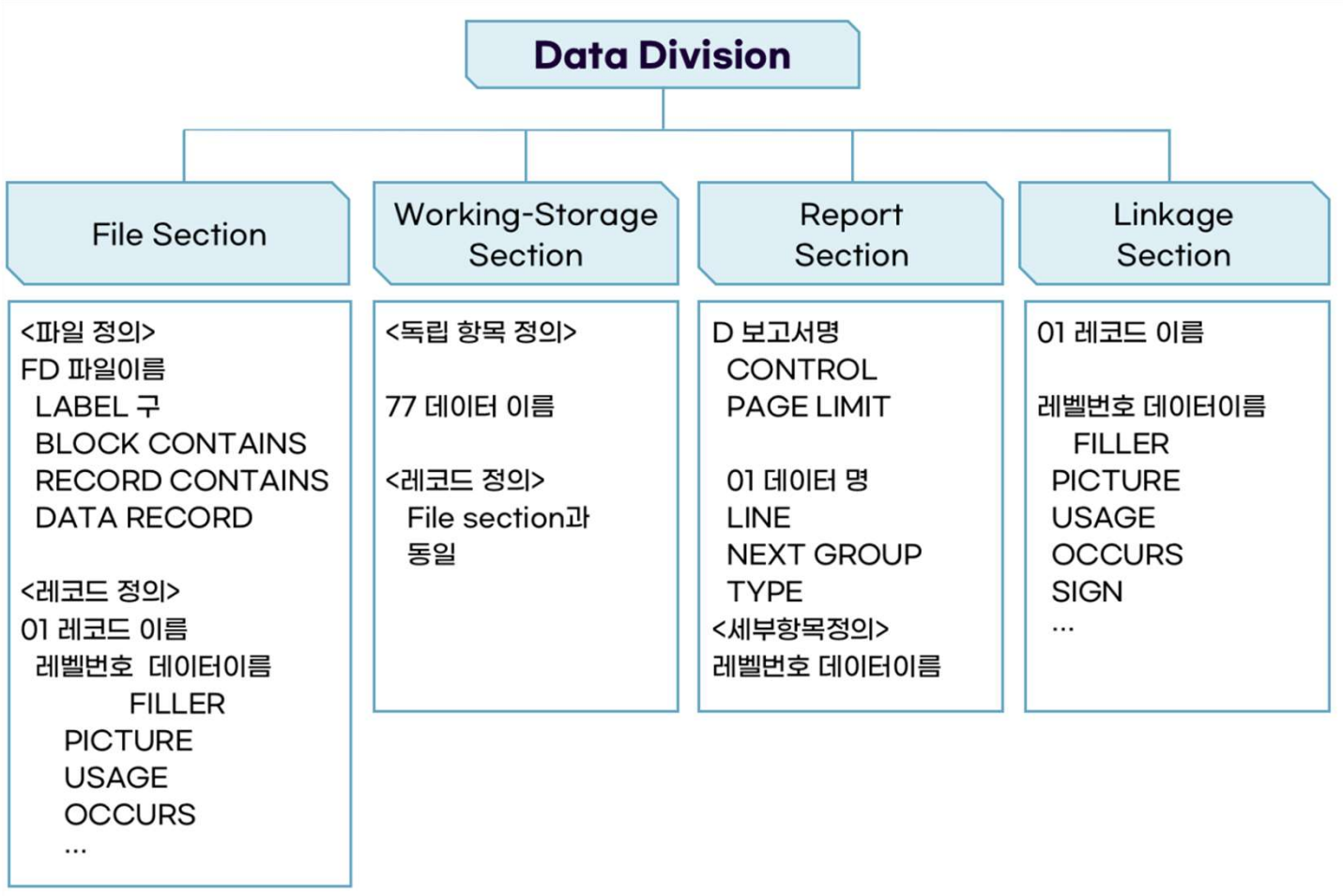
Paragraph (소절)

Sentence (문장)

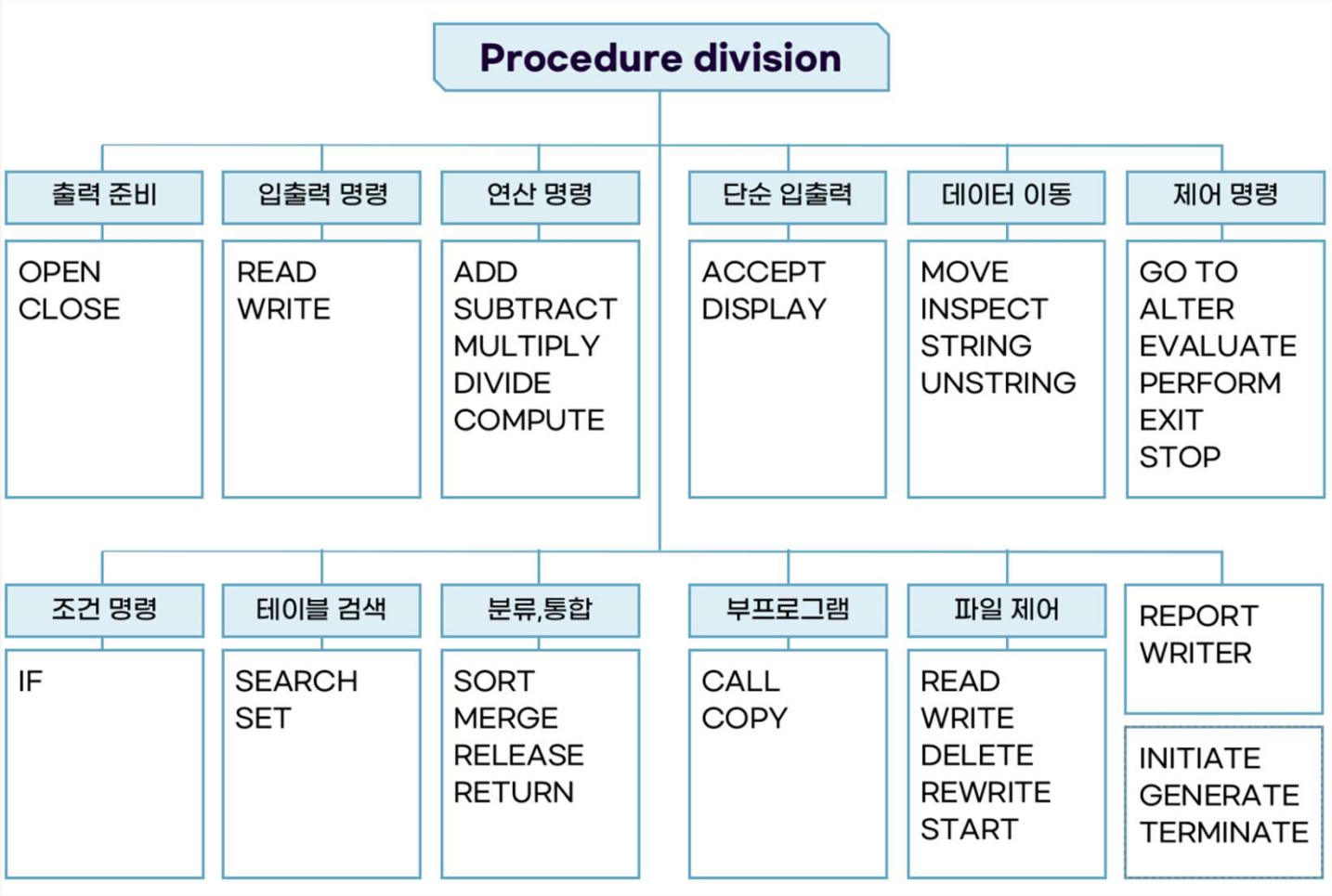
Cobol


Identification division
Environment division
Data division
Procedure division
Program-ID
Author
Installation
Data-written
Data-complied
Security
Configuration Section
Input-output Section
File section
Working-storage Section
Report section
Linkage section
입출력 준비 명령
입출력 명령
단말기 입출력 명령
데이터 이동 명령
제어 명령
조건 명령
테이블 검색 명령
...

Cobol


Data Division
File Section
Working-Storage Section
Report Section
Linkage Section
<파일 정의>
FD 파일이름
LABEL 구
BLOCK CONTAINS
RECORD CONTAINS
DATA RECORD
<레코드 정의>
01 레코드 이름
레벨번호 데이터이름
FILLER
PICTURE
USAGE
OCCURS
...
<독립 항목 정의>
77 데이터 이름
<레코드 정의>
File section과
동일
D 보고서명
CONTROL
PAGE LIMIT
01 데이터 명
LINE
NEXT GROUP
TYPE
<세부항목정의>
레벨번호 데이터이름
01 레코드 이름
레벨번호 데이터이름
FILLER
PICTURE
USAGE
OCCURS
SIGN
...

Cobol


Procedure division
출력 준비
OPEN
CLOSE
입출력 명령
READ
WRITE
연산 명령
ADD
SUBTRACT
MULTIPLY
DIVIDE
COMPUTE
단순 입출력
ACCEPT
DISPLAY
데이터 이동
MOVE
INSPECT
STRING
UNSTRING
제어 명령
GO TO
ALTER
EVALUATE
PERFORM
EXIT
STOP
조건 명령
IF
테이블 검색
SEARCH
SET
분류,통합
SORT
MERGE
RELEASE
RETURN
부프로그램
CALL
COPY
파일 제어
READ
WRITE
DELETE
REWRITE
START
REPORT
WRITER
INITIATE
GENERATE
TERMINATE

Cobol

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Sample

```
IDENTIFICATION DIVISION.
PROGRAM-ID. PRICE-TOTAL.
AUTHOR. S-KIM.
ENVIRONMENT DIVISION.
CONFIGURATION SECTION.
SOURCE-COMPUTER. PDP.
OBJECT-COMPUTER. PDP.
INPUT-OUTPUT SECTION.
FILE-CONTROL.
 SELECT INP-DATA ASSIGN TO INPUT.
 SELECT RESULT-FILE ASSIGN TO OUTPUT.
```

Cobol

```
DATA DIVISION.
FILE SECTION.
FD INP-DATA LABEL RECORD IS OMITTED.
  01 ITEM-PRICE.
     02 ITEM PICTURE X(30).
     02 PRICE PICTURE 9999V99.
     02 FILLER PICTURE X(44).
FD RESULT-FILE LABEL RECORD IS OMITTED.
  01 RESULT-LINE PICTURE X(132).
```

Cobol

```
WORKING-STORAGE SECTION.
77 TOT PICTURE 9999999V99, VALUE 0,
        USAGE IS COMPUTATIONAL.
77 COUNT PICTURE 9999, VALUE 0,
        USAGE IS COMPUTATIONAL.
 01 SUM-LINE.
   02 FILLER VALUE ' SUM=' PICTURE X(12).
   02 SUM-OUT PICTURE $$,$$$,$$9.99.
   02 FILLER VALUE ' NO. OF ITEMS= ' PICTURE X(21).
   02 COUNT-OUT PICTURE ZZZ9.
01 ITEM-LINE.
   02 ITEM-OUT PICTURE X(30).
   02 PRICE-OUT PICTURE ZZZ9.99.
```

Cobol

```
PROCEDURE DIVISION.
START.
    OPEN INPUT INP-DATA AND OUTPUT RESULT-FILE.
READ-DATA.
    READ INP-DATA AT END GO TO PRINT-LINE.
    ADD PRICE TO TOT.
    ADD 1 TO COUNT.
    MOVE PRICE TO PRICE-OUT.
    MOVE ITEM TO ITEM-OUT.
    WRITE RESULT-LINE FROM ITEM-LINE.
    GO TO READ-DATA.
PRINT-LINE.
```

Cobol

```
MOVE TOT TO SUM-OUT.
MOVE COUNT TO COUNT-OUT.
WRITE RESULT-LINE FROM SUM-LINE.
CLOSE INP-DATA AND RESULT-FILE.
STOP RUN.
```

Algol

Algol

- Algorithmic language
- 1950년대 말 ~ 60년대 초
- 범용 (general-purpose) 프로그래밍 언어
- Mother language: PL/I, Simula 67, Pascal, C, ...

특징

- ❑ 블록 (block) 구조
- ❑ 매개변수/파라미터/인자 전달

Pascal

Pascal

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- 1971, Niklaus Wirth

특징

- ❖ 프로그래밍 교육용
- ❖ 블록 구조
- ❖ 단순 (simple)

Pascal

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Sample

```
program MAIN(INFILE, OUTFILE);
const
  SIZE = 99;
type
  VECT = array[1..SIZE] of real;
var
  INFILE, OUTFILE: text;
  A: VECT;
  J, K: integer;
function SUM(V: VECT; N: integer): real;
   var
      temp: real;
      I: integer;
```

Pascal

```
  begin
    temp:=0;
    for I:= 1 to N do
     temp:=temp+V[I];
   SUM := temp
  end;
begin { main }
 read(INFILE, K);
  for J:= 1 to K do
   begin
      read(INFILE, A[J]);
      write(OUTFILE, A[J])
    end;
    writeln(OUTFILE, 'Total= ', SUM(A, K))
end
```

C

C

- 1972
- AT&T Bell Lab.
- Dennis Ritchie, Ken Thompson
- UNIX 운영체제 (operating system) 개발에 이용됨

특징

- ❖ 단순, 간결
- ❖ 효율적 (빠른 실행)
- ❖ 포인터 데이터 유형

단점

- 낮은 가독성
- 약한 타입 검사 (loosely type checking)
- 데이터 캡슐화 (data encapsulation) 지원 않음

C++

C++

- AT&T Bell Lab.
- Bjarne Stroustrup
- Simula 67 + C + class

특징

객체지향 (object-oriented) 언어

데이터 캡슐화 지원

학습 정리

학습 정리

- Fortran: 최초
- Cobol: business oriented
- Algol: mother language
- Pascal: 교육용
- C: 매우 중요한 언어
- C++: C + 객체 지향 개념